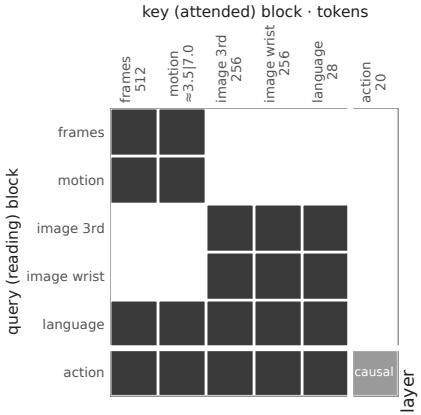
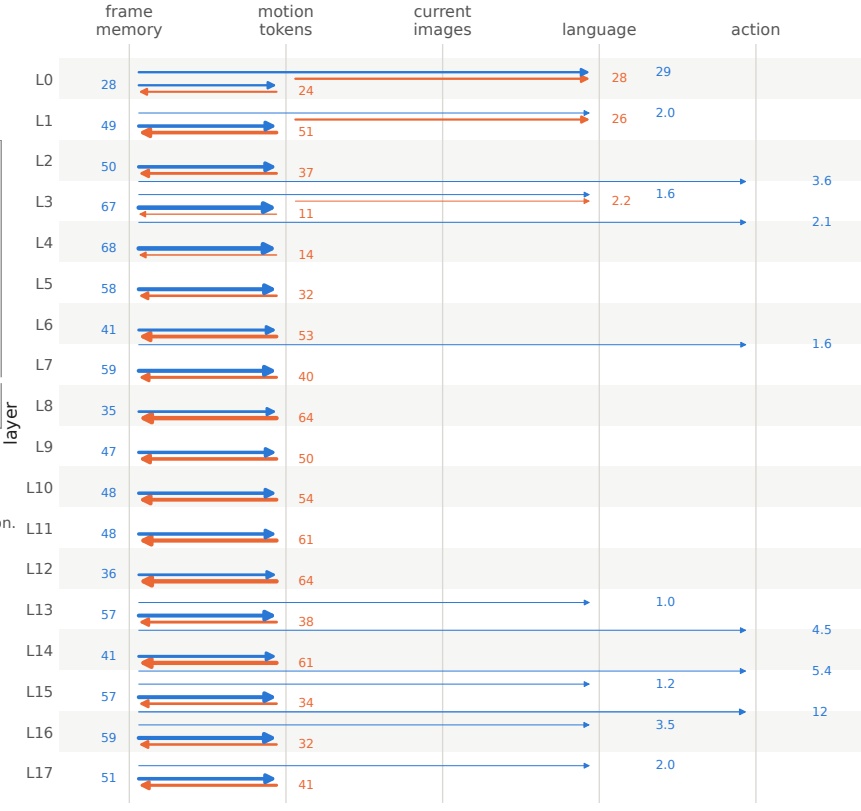
I. Two attention pathways carry motion-token content into the action tokens

Motion model, PatternLock | RouteStick, seed 42, 100 episodes / 495 inferences, head-averaged, episode-equal. (a) the attention mask as a block matrix and, per layer, how much each block reads from motion tokens and from frame memory; (b) the pathway edges per layer, both tasks pooled.

(a)



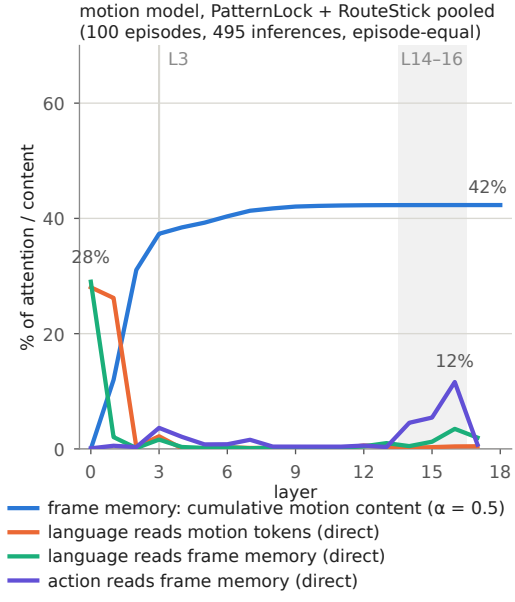
Block attention mask (dark = may attend; verified as exactly 0 elsewhere in every layer, head, episode). prefix = memory → images → language; suffix = action.



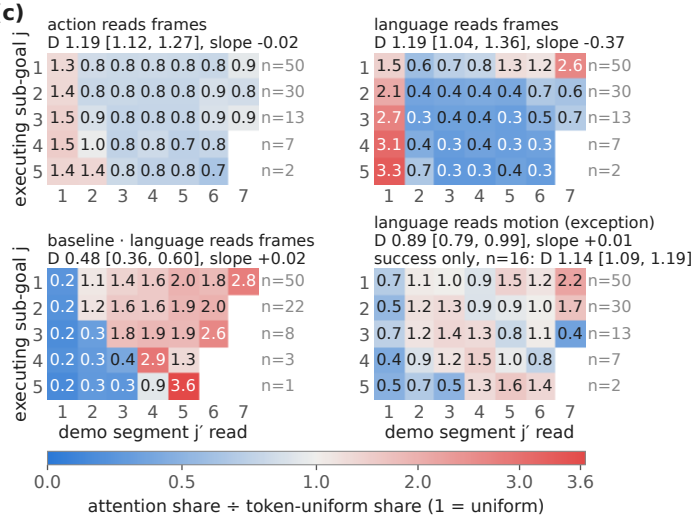
One row per layer. Only the six edges that carry memory content outwards are drawn: from motion tokens (orange) and from frame memory (blue) into motion / frames, language and action. Each horizontal arrow runs from the source to the reader; width and the number beside the reader's column = % of the reader's attention at that layer (edges below 1% not drawn). Two tasks pooled.

source: frame memory source: motion tokens

(b)



II. The pathways exist, but which demo segment is read shows no sub-goal alignment

RouteStick, motion model unless noted, 50 episodes per model, all inferences; rows pool episodes with different numbers of sub-goals m (n = episodes per row); frame stream weighted 'single', motion stream 'overlap'; D = diagonal enrichment, episode-level 95% CI.

(c)